# DHARANES CP

**Hardware Engineer**

Namakkal, Tamil Nadu | +91 7339355805 | mailtodharanes@gmail.com
LinkedIn



## PROFESSIONAL SUMMARY

Hardware-oriented Electronics and Communication Engineering undergraduate with hands-on internship experience in circuit design, microcontroller programming. Skilled in embedded systems, CST Studio-based signal and RF analysis for antenna design projects. Strong programming foundation in C++ and Java, with a problem-solving mindset, effective teamwork, and a strong eagerness to learn and apply engineering principles to real-world hardware solutions.

## EDUCATION

**B.E. / B.Tech, Electronics and Communication Engineering** 08/2023 – 2027
Kongunadu College of Engineering and Technology, Trichy | **CGPA:** 6.45

**Higher Secondary (12th), Malar Matric Hr. Sec. School** 2023
Paramathi, Namakkal

## INTERNSHIP EXPERIENCE

**Hardware Intern — Sam Core Solution Pvt Ltd, Trichy** 06/2025 – 07/2025

- Assisted in end-to-end circuit design and hardware testing for engineering projects.
- Programmed and debugged microcontrollers to support embedded hardware functionality.
- Debugged and validated PCB layouts, identifying and resolving hardware-level issues.
- Performed hardware validation and troubleshooting to ensure reliability and functional accuracy.

## PROJECTS

**Microstrip Patch Antenna and Wi-Fi/WiMAX Module** 05/2026
**Domain:** Communication | **Tools:** CST Studio

- Designed a compact, low-cost Microstrip Patch Antenna for Wi-Fi and WiMAX communication modules.
- Engineered the antenna to operate within required frequency bands using CST Studio-based analysis.
- Achieved efficient radiation, improved bandwidth, and reliable wireless connectivity.

**Microstrip Patch Antenna for Radar Application** 11/2025
**Domain:** Communication | **Tools:** CST Studio

- Designed and analyzed a low-profile, lightweight Microstrip Patch Antenna for radar applications in microwave frequency bands.
- Enhanced radar system performance through stable radiation characteristics and improved gain.
- Enabled reliable target detection via optimized antenna geometry and signal analysis in CST Studio.

## TECHNICAL SKILLS

**Programming Languages:** C++, Java
**Engineering & Simulation Tools:** CST Studio
**Core Domains:** Embedded Systems, Circuit Design, Antenna & RF Design, Signal Analysis
**Soft Skills:** Problem-Solving, Teamwork, Hardware Troubleshooting, Fast Learner

## COURSES & CERTIFICATIONS

- 24-Hour Hackathon — Schwan Tech 2.0
- NPTEL—Introduction to Industry 4.0 and Industrial Internet of Things
- Embedded Systems Course — Advanced Embedded System Application and IoT Programming (2025)
- JavaScript Course — Yellomatics, Trichy (2025): Syntax, functions, event handling, and client-side scripting.

## LANGUAGES

- Tamil
- English

## INTERESTS

- RF and Microwave Engineering
- Signal Processing
- IoT (Internet of Things)